and ray parenchyma. They are present in Pteridophytes,Gymnosperms and Dicots.

Phloem Fibres (or) Bast Fibres

The fibres of sclerenchyma associated with phloem are called phloem fibres or bast fibres. They are narrow, vertically elongated cells with very thick walls and a small lumen. Among the four phloem elements, phloem fibres are the only dead tissue. These are the strengthening as well as supporting cells.

Concept Map

Plant tissues

Meristematic tissue:
Capable of active cell division.
Thin walled and living.
Compactly arranged.
Found in root and shoot apex.

Based on position:
1. Apical
2. Intercalary
3. Lateral

Based on origin:
1. Primary
2. Secondary

Based on function:
1. Periderm (Epidermis)
2. Procambium (Primary vascular tissues)
3. Ground meristem (Cortex and Pith)

Based on division:
1. Mass meristem: Divides in all planes
2. Rib meristem: Anticlinal division in one plane
3. Plate meristem: Anticlinal division in two planes.

Permanent tissues:
Lose the power of cell division.
Have definite shape, size and form.

Simple tissues:
One type of cells.

Parenchyma:
Thin walled,isodiametric, found in all the parts.
Types:
1. Aerenchyma.
2. Storage parenchyma.
3. Stellate parenchyma.
4. Chlorenchyma.
5. Prosenchyma.

Collenchyma:
Hypodermal position. Provide mechanical strength.
Types:
1. Angular collenchyma.
2. Lacunar collenchyma.
3. Lamellar collenchyma.

Sclerenchyma: Dead cells and lignified walls.
Types:
1. Sclereids
2. Fibres

Sclereids:
1. Brachysclereids: Stone cells
2. Macro sclereids: Rod shaped
3. Osteosclereids: Bone shaped
4. Astrosclereids: Star shaped
5. Trichosclereids: Hair cells

Fibres:
1. Wood fibres: xylary fibres
2. Bastfibres: Extra xylary fibres
3. Surface fibres: *Cottan*
4. Mesocarp fibres: *Coconut*
5. Leaf fibres: *Agave*

Complex tissues:
More than one type of cells.

Xylem: Water conducting tissue.
1. Tracheids: Dead, elongated with tapering end
2. Vessels: Made of row of dead cells
3. Xylem fibres: Lignified and sclerenchymatous.
4. Xylem parenchyma: Living and cellulosic

Phloem: Food conducting tissue
1. Sieve elements: Sieve cells & sieve tubes
2. Companion cells: Only in Angiosperms.
3. Phloem parenchyma: Living & absent in Monocots.
4. Phloem fibres: Thick walled & sclerenchymatous, giving mechanical strength.



Syncyte: Cell which is formed by fusion of cells is called Syncyte.

Example: Vessels (Dead syncyte), sieve tube (living syncyte)

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